

Hansol Ryu

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Education

Seoul National University (SNU)

B.S. Candidate in Mechanical Engineering

Double Major in Computer Science and Engineering

- GPA: **4.19 / 4.30**

Seoul, Korea

Jan. 2021 – Present

Research Interests

Robot Learning, Reinforcement Learning, Imitation Learning, Learning-Based Control, and Contact-Rich and Dexterous Manipulation

Publications

- S. H. Jeong, M. Kim, **H. Ryu**, J. Noh, and A. Kim*, “Post-Grasp Retrieval in Cluttered Scenes via Zero-Shot VLM-Guided Spatial Reasoning and Active Perception,” *IEEE Robotics and Automation Letters (RA-L)*, under review, 2026.

* Corresponding author.

Research Experience

Research Intern (B.S. Thesis), Robotics Laboratory

Advisor: Prof. Frank C. Park, Seoul National University

Seoul, Korea

Mar. 2026 – Present

- **B.S. Thesis:** Visuomotor policies for contact-rich manipulation, focusing on geometric representations, policy equivariance, and feedback-based robustness.
- Investigated representation choices related to policy equivariance, comparing world-frame and end-effector-local representations under geometric transformations.
- Developed simulation and evaluation pipelines combining diffusion-based visuomotor policies with Cartesian impedance/compliance control and force-feedback regulation.

Research Intern (UROP), Robust Perception and Mobile Robotics Lab (RPM Robotics Lab)

Advisor: Prof. Ayoungh Kim, Seoul National University

Seoul, Korea

Jan. 2026 – Jul. 2026

- Contributed to a VLM-guided active-perception framework for disturbance-aware post-grasp retrieval in cluttered scenes, including implementation and evaluation of a TSDF-based active-perception baseline and investigation of TSDF and 3D Gaussian Splatting as scene representations.
- Developed a VLM-based retrieval-direction pipeline that evaluates candidate extraction directions from visual observations and maps the selected direction to executable robot motions.
- Designed and implemented the real-robot evaluation pipeline and quantitative metrics for retrieval success and collateral scene disturbance.

Research Intern, Interactive and Networked Robotics Laboratory (INRoL)

Advisor: Prof. Dongjun Lee, Seoul National University

Seoul, Korea

Sep. 2025 – Mar. 2026

- Investigated numerical methods for contact dynamics based on relaxed nonlinear complementarity formulations.
- Studied the integration of contact solvers with model predictive control for physically grounded robot motion generation and control.
- Evaluated learned locomotion policies under different contact-modeling and simulation settings.

Relevant Coursework

- **Optimization and Machine Learning** – A+
Topics: optimization on manifolds, optimal control, reinforcement learning
- **Basics of Deep Learning** – A+ (**1st in class**)
- **Introduction to Robotics** – A+
- **Dynamics** – A+
- **Mechanical System Modeling and Control** – A0
- **CS285: Deep Reinforcement Learning, UC Berkeley** – self-study, ongoing
- **Stochastic Control and Reinforcement Learning** – planned for Fall 2026

Selected Projects

- LiDAR-Based Noninvasive Sleep Monitoring and Environment-Adaptive IoT Hub** *Fall 2025*
- Developed a noninvasive sleep-monitoring system using an 8×8 time-of-flight LiDAR array and multimodal environmental sensors.

Awards and Honors

- Mechatronics Grand Prize (1st Place)** *Fall 2025*
Seoul National University
- National Science & Technology Scholarship** *Jan. 2023 – Present*
Korea Student Aid Foundation (KOSAF)
- Scholarship for Academic Excellence** *Jan. 2022; renewed Jan. 2023*
Seoul National University

Military Service

- Korean Augmentation to the U.S. Army (KATUSA)** *18 months*
Republic of Korea Army / U.S. Army
- Served in a U.S. Army unit in an English-speaking operational environment; received the U.S. Army Commendation Medal (ARCOM).

Skills

- Programming:** Python, C++, Java, MATLAB
- Machine Learning:** PyTorch, NumPy, diffusion-based visuomotor policies
- Robotics:** ROS 2, MuJoCo, robot dynamics, impedance/compliance control, contact-rich simulation
- English:** TOEFL iBT 110/120 (5.5/6.0, new scale); TEPS 488/600